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End-of-surgery prediction of postoperative infectious complications from intraoperative vital-sign dynamics

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Abstract

Infections after surgery remain a leading cause of morbidity and mortality, yet reliable risk stratification at the end of surgery is limited. Intraoperative vital signs are continuously recorded in modern operating rooms but remain an underexploited source of real-time prognostic information. We developed and validated a machine-learning model integrating intraoperative vital-sign dynamics to predict postoperative infections immediately at the end of surgery. We extracted arterial blood pressure, heart rate, oxygen saturation, temperature, and end-tidal CO₂ time-series from a clinical data warehouse, transforming these signals into interpretable summary, trend, and distributional descriptors. Using routine data from 10,719 surgical procedures, models incorporating interpretable intraoperative time-series features achieved an AUROC of 0.88 (95% CI, 0.85–0.91) for infection prediction at the end of surgery, significantly outperforming models based on preoperative variables alone. Model predictions were calibrated across major procedure clusters and interpretable through SHAP-based feature attribution. Our results demonstrate that intraoperative time-series data encode signatures of cumulative surgical and physiological stress, revealing early and clinically actionable signals of postoperative infection risk and enable an explainable machine-learning framework for perioperative monitoring systems. This explainable approach moves risk assessment from delayed postoperative testing to immediate, digital decision support, ready for integration into perioperative monitoring systems.

Introduction

Postoperative infections such as surgical site infections (SSIs), pneumonia, urinary tract infections, or sepsis, are among the most frequent and significant surgical complications^{1,2}. Their average incidence ranges from 0.4% to 5% and is associated with higher morbidity, prolonged hospital stays, and increased healthcare costs, making early identification of patients at risk crucial for timely interventions^{3–6}. Numerous factors predispose patients to postoperative infection. Besides the key predictors surgical type, contamination, operative duration and obesity, relevant contributors include glycaemic control, smoking, immunosuppression, and the timing and appropriateness of antibiotic prophylaxis^{7,8}. These factors have been incorporated into existing logistic regression-based risk models (e.g. the ACS-NSQIP risk calculator)^{9–11}. More recently, machine-learning (ML) approaches have been applied to model postoperative complications, including bacterial infections, and have demonstrated improved predictive performance compared with traditional methods^{12–14}. ML-based prediction of pneumonia^{13,15,16}, SSI^{17–21}, sepsis^{22–24} or composite endpoints^{25–27} has achieved moderate to good discrimination¹⁷. These studies, however, relied on static preoperative factors such as emergency status, ASA score, procedure type. However, perioperative risk is not solely determined by fixed baseline characteristics; it evolves dynamically during surgery. Intraoperative physiology is captured at high frequency in modern operating rooms, yet its full temporal structure is rarely exploited for risk modelling. As a result, potential digital physiologic “signatures” of downstream complications embedded in vital-sign dynamics remain underused.

End-of-surgery risk stratification is a natural decision point at which preoperative knowledge can be enriched by intraoperative events to guide early postoperative management²⁸. Risk calculated immediately at the end of surgery, defined as skin suture, has the potential to create a distinct timepoint for “postoperative” prediction enabling timely interventions such as targeted monitoring or preventive bundles^{28,29}. Indeed, machine-learning models using intraoperative data in addition to preoperative factors resulted in improved prediction of postoperative complications^{13,14}. However, in these previous models, the intraoperative physiology was reduced to conventionally summary or pointwise measures (e.g., means, extrema, or last recorded values). While some recent studies have begun leveraging temporal dynamics³⁰, most approaches still do not exploit the full higher-order distributional structure of continuous intraoperative physiological time-series data^{28,31}. This limitation is largely driven by the intrinsic complexity of continuous intraoperative data, characterized by irregular acquisition frequencies, variable signal duration, and substantial

measurement noise, which has hindered the extraction of robust, clinically meaningful, predictive features³².

Our overarching goal was to incorporate the full information captured in intraoperative time-series data to deliver an accurate, interpretable infection risk score immediately at the end of surgery that can be computed in real time within existing monitoring systems. Using real-world electronic health record (EHR) data, we extracted intraoperative vital-sign time-series and tested whether such data streams improve early prediction of postoperative infections beyond established risk factors. To overcome the limitations of prior ML models relying on static variables^{28,31}, we transformed full intraoperative signals into statistical, trend, and higher-order features (e.g., skewness, kurtosis, entropy), allowing to capture temporal dynamics and physiological instability²⁸.

Results

Study cohort characteristics

The total surgeries included in the study were 15,221 surgery cases (Supplementary Figure 1). Excluding surgeries with pre-existing infections or missing/inconsistent present-on-admission (POA) data resulted in cohort of 10,719 surgeries with a median surgery duration of 3 h and 33 min (IQR, 2 h 27 min – 4 h 41 min). Postoperative bacterial infection not present at admission occurred in 504 surgeries (4.7%). Bacterial infection incidence varied across the surgical departments, ranging from 2.3% in thoracic surgeries to 11.6% in visceral surgeries. Infection rate, patient characteristics and operative procedure details across the surgical departments are presented in Supplementary Table 1. Evaluation of the intraoperative vital-sign monitoring practices over the 8-year study period showed that the availability of most continuous streams remained stable. However, there was a slight decline in 2020 before recovering in 2021, likely reflecting shifts in clinical practice and airway management protocols during the COVID-19 pandemic (Supplementary Figure 2). Because inclusion required complete high-fidelity monitoring, the modelled cohort represented a population with higher clinical acuity and surgical complexity compared to excluded cases (Supplementary Table 2). Specifically, included patients were older (median age 66 vs. 61 years, SMD 0.312), exhibited higher baseline ASA scores (SMD 0.572), and underwent significantly longer operations (median 213 vs. 88 minutes, SMD 0.970). The procedure mix differed substantially, with a higher concentration of complex surgeries that routinely necessitate invasive arterial lines and continuous monitoring (cardiac, vascular, and thoracic procedures). Consequently, the excluded cohort had a significantly lower postoperative infection rate than the included cohort (2.8% vs. 4.7%).

Clinical features from the EHR were grouped by timing: preoperative, intraoperative and postoperative. End-of-surgery prediction at skin suture used preoperative and intraoperative variables (time-series features and surgical factors: Figure 1).

Model performance across the surgical timeline

We developed three models aligned with the surgical timeline: a start-of-surgery model (M1), an end-of-surgery model (M2), and an early postoperative model (M3) (Figure 2). M1 represents the baseline and includes preoperative variables age, sex, ASA score, comorbidities, preoperative antibiotics use, emergency surgery and night surgery. The AUROC of M1 to predict postoperative infections was 0.806 [95% confidence interval (CI): 0.786 - 0.827]. Across different ML model

architectures, the Random Forest model provided the best overall discriminatory power (Supplementary Figure 3). For M2, intraoperative variables available at the end of surgery were added. Discrimination at M2 was significantly improved with an AUROC of 0.880 [95% CI: 0.850 - 0.911]. M3 further incorporates laboratory values obtained within 48 hours (POD2). Discrimination at M3 was significantly increased with an AUROC of 0.903 [95% CI: 0.889 - 0.917]. Positive predictive value and sensitivity increased from M1 to M3, while specificity remained high across all models. To evaluate the potential of our approach in settings without invasive monitoring, we additionally trained a reduced-feature model (M2-min) excluding arterial blood pressure metrics and end-tidal CO₂, relying solely on SpO₂, temperature, and heart rate dynamics. This minimal model maintained high discrimination with an AUROC of approximately 0.87, compared to 0.880 for the full M2 model, demonstrating the robustness of basic non-invasive vital-sign time-series for infection prediction (Supplementary Figure 4). These findings suggest that similar physiologic signatures may also be detectable in lower-acuity procedures where invasive monitoring is not routinely used, extending the potential applicability of this approach to a broader range of surgical settings.

Relative contribution of patient, procedural and physiological parameters

To disentangle the sources of predictive signal at the end of surgery, we quantified the relative contribution of patient, procedural, and intraoperative physiological parameters to end-of-surgery infection prediction (M2) using grouped mean absolute SHAP values (Figure 3a).

This analysis enables transparent assessment of how dynamic intraoperative physiology adds to baseline patient risk and procedural context in driving model predictions. Physiological time-series features accounted for 44.4% of the total attribution, exceeding patient factors (42.1%) and procedural variables (13.5%) (Figure 3b). Within the physiological contribution, summary statistics contributed 29.5%, trend features 9.3%, and higher-order statistics 5.6%. Furthermore, the model captured distinct non-linear trends in the intraoperative physiological data, where the risk contributions fluctuated across value ranges rather than following a simple linear relationship (Figure 3c).

Key physiologic parameters of postoperative infection prediction

Having established that intraoperative physiological parameters contribute substantially to end-of-surgery infection prediction, we next examined which specific physiologic signals and time-

series characteristics were most strongly associated with postoperative infection risk. The predictive relevance of intraoperative physiologic variables was identified by absolute mean SHAP value and grouped into summary, trend, and higher-order feature categories (Figure 4a).

Detailed examination of the feature importance of physiological variable revealed distinct risk profiles across statistical categories (Figure 4b–d). Within the **summary statistics**, high oxygenation (high median and max SpO₂) and lower minimal heart rates were protective factors, whereas elevated temperature (mean and minimum T_{n/o}) were associated with elevated risk of postoperative infection (Figure 4b). **Trend statistics** localized the risks to higher T_{n/o} baseline and higher heart rate baseline as risk factors, while a higher SpO₂ baseline remained protective (Figure 4c). The models **higher-order statistics** captured the impact of physiological instability, for instance, reduced entropy and kurtosis in blood pressure metrics (MAP, DBP) were protective, whereas increased SBP entropy and MAP kurtosis signaled elevated risk (Figure 4d). Collectively, while summary statistics contributed the largest overall share of physiological attribution, trend and higher-order features provided complementary signal capturing dynamic and distributional instability patterns (Figures 3b and 4a). To assess how the feature dependencies translated into measurable gains in predictive performance, we performed a systematic feature-group ablation analysis. In the ablation analysis, discrimination at end-of-surgery was driven primarily by trend-derived features (highest AUROC observed for models including trend features), whereas adding higher-order statistics altered AUROC only marginally. Higher-order statistics provided at most a modest incremental contribution, most noticeably improving precision (PPV) when added on top of trend features (Supplementary Figure 5).

Association of intraoperative physiology with prolonged surgical duration

To examine which intraoperative dynamics track prolonged operations, we modelled surgical duration as quartiles (Q1–Q4) and tested which factors shift cases toward the longest-duration quartile (Q4). After reducing redundancy among highly correlated predictors, selecting informative features using LASSO, and fitting an adjusted ordinal regression model with BH-FDR correction, the strongest associations were observed for vital-sign dynamics reflecting intraoperative instability and extremes. In particular, diastolic blood pressure variability, low minimum diastolic blood pressure, and low minimum ventilatory CO₂ were among the largest adjusted effects, each associated with a substantially increased probability of belonging to Q4 (Supplementary Figure 6). For example, a 1-SD increase in diastolic BP variability was associated with OR 8.15 (95% CI 6.96–

9.53; $q=3.12 \times 10^{-149}$), corresponding to an absolute $\Delta Pr(Q4)$ of +30.6 percentage points. Minimum diastolic BP (OR 4.87, 4.26–5.57; $q=5.14 \times 10^{-118}$) and minimum ventilatory CO_2 (OR 4.12, 3.50–4.85; $q=1.07 \times 10^{-63}$) showed similarly large probability shifts ($\Delta Pr(Q4)=+19.5$ and +16.4 percentage points, respectively).

Model calibration and sensitivity analysis

A subgroup analysis was conducted to assess biases across the different procedure type clusters. Cluster-level calibration bias was slightly positive for all procedure clusters, indicating a systematic overestimation of bacterial infection risk (Supplementary Figure 7a). Brier scores ranged from 0.02 to 0.08 in most clusters, indicating high predictive accuracy, exceptions were gastrointestinal, and cutaneous procedures, which had higher scores (0.10 and 0.12, respectively). For reference, a Brier score of 0.25 corresponds to random predictions when outcome incidence is ~50%. Calibration curves confirmed these patterns, showing a monotonic trend of overestimation that was most pronounced in cardiac and endocrine clusters, and conversely, abdominal wall and cutaneous procedures demonstrated the highest fidelity to the ideal calibration (diagonal line), particularly at higher risk strata (Supplementary Figure 7b).

Discussion

To our knowledge, this is the first study to predict a broad spectrum of postoperative bacterial infections at the moment surgery ends using full intraoperative time-series combined with preoperative risk factors. Prior work relied on static preoperative predictors or on postoperative data leaving the interval between skin closure and the first postoperative assessments underexplored^{13,14,27,31}. By shifting prediction into this window, our approach provides actionable information at this moment when no postoperative laboratory values are yet available but key decisions on treatment must be made.

Based on the hypothesis that intraoperative physiologic instability serves as a proxy for cumulative physiologic stress, patient vulnerability, or surgical complexity associated with downstream infection risk, this study tested whether intraoperative vital-sign time-series data enable accurate prediction at the end of surgery²⁹. The performance gain from preoperative to end-of-surgery models exceeded that from end-of-surgery to POD2, indicating that most predictive information is captured intraoperatively. This may be driven by the high information density and measurement frequency of the intraoperative data compared to the relatively sparse nature of postoperative

laboratory values. Beyond known deviations in summary and trend measures¹³, higher-order features (kurtosis, skewness, entropy) provided complementary signal capturing distributional instability and complexity patterns not accessible through conventional summary metrics. Systematic ablation analysis revealed that overall discrimination was driven primarily by trend-derived features, whereas higher-order statistics added at most a modest incremental contribution, most noticeably improving precision when combined with trend features. Together, these findings support dynamic perioperative risk modelling that integrates both physiologic trajectory and signal complexity.

Among physiologic predictors, elevated heart rate and $T_{n/o}$ baseline were associated with higher risk of infection, while higher SpO₂ baseline remained protective, reflecting early deviations in thermoregulation and ventilation associated with adverse outcomes^{33–38}. Higher-order statistics captured distinct aspects of physiological instability: entropy reflects unpredictability; for example, increased SBP entropy reflects irregular and less tightly regulated systolic blood pressure, suggesting reduced homeostatic control. Kurtosis highlights the presence of extreme, intermittent deviations, e.g. increased MAP kurtosis reflects occasional pronounced peaks or drops in mean arterial pressure rather than smooth variability around a stable mean. These mathematically derived features, however, must be interpreted cautiously, as they are intrinsically more susceptible to transient monitoring artifacts or surgical manipulations than standard summary metrics. Together, these findings suggest that dynamic variability, asymmetry, and instability within intraoperative vital signs are associated with physiologic stress and higher risk of infection. Consistent with this interpretation, ordinal duration-quartile modelling showed that intraoperative instability and extremes (e.g., blood pressure variability and minima, and low ventilatory CO₂ minima) were strongly associated with prolonged operations even after adjustment for baseline EHR covariates and procedure category. This suggests that high-frequency intraoperative dynamics act as proxy indicators of cumulative surgical and physiologic stress burden, reflecting surgical complexity and patient vulnerabilities that are associated with infection risk.

A key strength of our study is integrating intraoperative dynamics with machine learning and XAI to enable end-of-surgery prediction. From a technical perspective, we implement a feature-engineering pipeline that converts dense intraoperative time-series into summary, variability, and higher-order statistics²⁸, enabling dynamic risk modelling without requiring deep learning architectures or raw waveform access. This design mirrors recent work demonstrating that statistical features derived from short- and long-term variability are key inputs for perioperative risk estimation²⁸ and that combining fixed and time-series physiologic measurements improves prediction of

major postoperative complications³⁹. We leverage SHAP to break down each prediction into feature contributions, which is important in approaching a clinical adoption with an interpretable model that highlights the added risk based on intraoperative measurements.

The clinical translation of this study will result in a robust risk stratification already at the end of surgery, creating a clinically meaningful window for targeted postoperative surveillance, early diagnostics, and timely antimicrobial or interventional treatment^{28,29}. The fact that discrimination at the end of surgery (M2) approaches that of models incorporating postoperative laboratory data (M3) suggests that the intraoperative period contains rich physiologic information about downstream infection risk that is not captured by standard postoperative markers alone. In practical terms, this shifts the prediction window forward by up to 48–72 hours compared with conventional laboratory-based risk assessment.

Given current evidence, a high end-of-surgery risk estimate should primarily guide surveillance. High-risk patients could be prioritized for intensified monitoring^{28,29}. By defining intraoperative physiologic signatures associated with infection risk, our work provides the foundation for future multicentre studies to test specific management strategies triggered by these risk estimates. This aligns with the broader vision of intraoperative and immediate postoperative risk models²⁹ and enables proactive instead of reactive infection management without additional workflow burden.

There are a few limitations in our study. Requiring complete intraoperative vital-sign traces for SBP, DBP, MAP, heart rate, SpO₂, T_{n/o} and EtCO₂ introduces selection bias (Supplementary Table 2). Patients requiring such comprehensive invasive monitoring inherently represent a population with higher comorbidities and surgical complexity and therefore also infection rates. Because the use of invasive monitoring is clinically indicated rather than random, the resulting selection bias reflects confounding by indication and cannot be fully resolved using retrospective routine data. Future studies may partially address this through multicentre datasets including different monitoring practices, prospective study designs, and causal inference approaches to capture the clinical factors guiding invasive monitoring decisions and to enable a more specific assessment of both generalizability and monitoring-related selection mechanisms. Crucially, our model does not account for intraoperative interventions such as vasopressor administration, fluid resuscitation, or anaesthetic titrations. Consequently, the observed vital-sign dynamics reflect the net physiological state resulting from both underlying surgical stress and active anaesthesia management, highlighting the need for future studies that jointly model physiologic signals and intraoperative interventions allowing to assess the impact of specific management strategies. The model predicts

infections as an aggregate rather than by specific type (e.g., pneumonia vs. sepsis vs urinary tract infection) allowing to provide a broad “early warning” signal for systemic vulnerability. Future studies that need to include even larger, multicentric datasets will allow subtype-specific analyses to tailor interventions precisely to the type of predicted infection. While formal wound classification (clean, clean-contaminated, contaminated, dirty) was not consistently available, procedure cluster and emergency status at least in part reflect contamination burden. Furthermore, a sensitivity analysis evaluating the explicit inclusion of uncomplicated and complicated diabetes diagnoses revealed them to have minimal predictive impact at the end of surgery (ranking 52nd and 155th in feature importance, respectively). Future studies incorporating more granular preoperative variables, such as smoking status, dynamic measures of glycaemic control (e.g., HbA1c), and precise documentation of antibiotic prophylaxis timing, may further refine perioperative risk modelling. While the model demonstrated high predictive accuracy, a systematic overestimation of infection risk across procedure clusters suggests that clinical implementation may require the calibration of decision thresholds to mitigate potential alarm fatigue. Adjusting these thresholds based on specific cluster-level biases would refine the balance between early detection and the practicalities of clinical postoperative care. Future work will evaluate subtype-specific models or subtype-aware recalibration as larger cohorts become available. Finally, while SHAP improves interpretability, feature attributions remain *post hoc* and may not reflect causal mechanisms. Furthermore, this study requires validation in other healthcare environments.

Postoperative infections remain a major source of patient harm and cost, and early prediction, especially at the moment surgery ends, is currently limited^{13,40}. Our findings show that intraoperative vital-sign dynamics contain early predictive signals and that combining them with preoperative factors yields high discrimination and interpretable explanations. This approach may enable intelligent monitoring systems in the operating room and recovery unit that provide early forecasts of downstream complications. Future work will evaluate prospective deployment and external validation.

Methods

Study design

This retrospective study analysed EHR data from surgical procedures performed at University Hospital in Bern between May 2014 and September 2022. The study was approved by the Cantonal Ethics Committee of Bern (KEK-ID: 2021-00965), with a waiver of informed consent due to

the study's retrospective design. An initial dataset of 64,942 surgical cases across seven departments (vascular, orthopaedic, visceral, thoracic, gynaecology, otolaryngology (ENT), and paediatric) has been described in detail elsewhere⁴¹. The analysis took place from October 2024 to August 2025. All statistical analyses were conducted using Python (3.11.10).

Cohort size and exclusions

The main reduction from the initial dataset (64,942 cases) via the study cohort (15,221) to the final modelling cohort (10,719 cases) was driven by the requirement for complete intraoperative monitoring of the selected vital-sign streams (SBP, DBP, MAP, heart rate, SpO₂, nasopharyngeal temperature, and EtCO₂). Because time-series features were computed separately for each vital sign and we did not impute missing intraoperative signals, cases lacking one or more of these streams (e.g., no arterial line, missing EtCO₂ or temperature monitoring, or insufficient signal coverage after 5-min resampling) were excluded. To ensure robust modelling of high-resolution physiologic time-series data, we required complete intraoperative monitoring streams. To evaluate potential selection effects associated with this requirement, baseline characteristics, procedure mix, and infection rates were compared between the included and excluded cohorts (Supplementary Table 2). In addition, model inclusion required linkage to structured EHR variables used in modelling (e.g., procedure coding and diagnosis coding/endpoint ascertainment), which was unavailable for a subset of cases (Supplementary Figure 1).

Endpoint and handling of patients with infections present on admission

Bacterial infections were identified and the EHR data manually annotated based on a list (Supplementary Table 3) of International Statistical Classification of Diseases, Tenth Revision codes (ICD-10 codes, German modification). This analysis included surgical site infections (superficial incisional/skin-soft-tissue and deep incisional/organ-space), pneumonia, upper-airway/ENT, central nervous system, gastrointestinal, genitourinary, sepsis, and other specific bacterial infections. Postoperative infection was defined as any ICD-10 bacterial infection diagnosis not flagged as present on admission (NPOA). This composite endpoint was chosen to maximize statistical power at a single end-of-surgery prediction timepoint and to reflect the practical clinical task of identifying patients at elevated overall postoperative infectious risk. Accordingly, the model should be interpreted as predicting general postoperative infection risk rather than specific infection subtypes. Diagnoses flagged as POA were considered pre-existing and were not counted as postoperative outcomes. Cases with both POA and non-POA bacterial infection codes were labelled as 'mixed' and excluded to ensure unambiguous endpoint timing. Cases with POA-only infection codes (and no non-POA infection code) were treated as pre-existing infections and excluded.

Preoperative Patient Characteristics and Comorbidities

The EHR included patient's age, sex and assigned ASA score. Patient comorbidities, present in patients upon admission, were derived from ICD-10-GM codes using the Charlson Comorbidity Index⁴² and the Elixhauser Comorbidity Score⁴³. Presence of comorbidities was encoded as binary indicators using Charlson and Elixhauser definitions.

Intraoperative and Procedural Variables

In the EHR, procedures were encoded using the Swiss operation procedure classification scheme (CHOP) provided by the Federal Statistical Office⁴⁴. These CHOP-coded procedures were transformed into a more comparative scheme using the validated multi-embedding framework MAP-CARE⁴⁵. Codes were translated and enriched along clinically relevant domains, including anatomy, action, organ system, and invasiveness. Clusters of procedure types were finally reviewed and clinically validated by an experienced surgeon, resulting in 12 semantically distinct procedure clusters encompassing nervous system, endocrine, thoracic, cardiac, vascular, abdominal wall, gastrointestinal (upper and lower), hepatobiliary and pancreatic, urologic and male genital ("uro-andrological"), female genital, musculoskeletal extremity, and cutaneous procedures. For each surgical case, the number of CHOP codes per cluster was tallied, yielding one numeric feature per cluster (Supplementary Table 4). The administration of preoperative antibiotics was extracted from the EHR and included as a binary variable. Timing, agent appropriateness, and redosing were not consistently available and were therefore not modelled.

Extraction of Intraoperative Data

Intraoperative time-series data were extracted from the clinical data warehouse, including arterial blood pressure (systolic, diastolic, and mean arterial pressure; SBP, DBP, MAP), heart rate, peripheral oxygen saturation (SpO_2), nasopharyngeal temperature ($T_{n/o}$), end-tidal carbon dioxide (EtCO_2). All signals were resampled to 5-min intervals using mean values. Short internal gaps within otherwise available streams were linearly interpolated after resampling. This resampling of the signal harmonizes the time-series data with differing acquisition frequencies. To ensure data quality, vital-specific exclusion ranges (Supplementary Table 5) were defined by an experienced surgeon. Observations falling outside these ranges (e.g., systolic arterial blood pressures <50 or >300 mmHg; oxygen saturation $<70\%$) were treated as likely artifacts and replaced by the most recent valid measurement (last observation carried forward). We did not perform full-sequence imputation for absent intraoperative channels. Cases missing one or more required vital-sign streams, or with insufficient coverage after resampling, were excluded from model development.

Feature Engineering from Time-Series Data

Three types of statistical features were generated through time-series extraction (TSE) (Figure 1): 1. **Summary statistics:** Mean, median, standard deviation, minimum, and maximum were calculated for each signal. Local extrema were characterized by detecting peaks (positive events) and troughs (negative events) using vital-specific thresholds (e.g., 10% relative change for blood pressure and heart rate; 0.4 °C absolute change for temperature). From these detections, we computed the number of events and the average inter-event interval (IEI). 2. **Trend statistics** across the surgery were extracted by fitting a linear regression to each time-series, yielding slope (Trend over Time) and intercept coefficients (Baseline) for all parameters. 3. **Higher-order statistics:** Higher-order features captured the distributional and dynamic structure of intraoperative signals. Skewness quantified asymmetry in the signal distribution, indicating whether values tended to cluster above or below their physiological baseline. Kurtosis described the concentration of data around the mean and the frequency of extreme deviations — high kurtosis reflects intermittent sharp peaks or drops, suggesting physiological instability, whereas low kurtosis indicates steadier variability. Approximate entropy, a nonlinear metric of signal irregularity and unpredictability, quantified the degree of physiological control — high entropy reflects more variable and less predictable local dynamics, while low entropy represents stable, well-regulated vital-sign patterns⁴⁶. The signal burden, calculated as the area under the curve (AUC) using Simpson's rule, summarized the cumulative magnitude and duration of physiologic deviation across the entire operation. Exemplary traces to showcase higher-order statistics are shown in Supplementary Figure 8.

Postoperative Variables and Handling of Missing Data

To compare predictive performance at the end of surgery with that at later postoperative time points, we incorporated laboratory measurements collected on postoperative days 0 to 2 (POD0–2) (Supplementary Table 6). Missing data were addressed through imputation. For each POD and marker, data imputation was performed by training an histogram-based gradient boosting (HGBoost) algorithm, as we described before⁴¹.

Model Development

We followed the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) reporting guideline for model development and validation⁴⁷. We used Logistic Regression, Random Forest and Gradient Boosting (XGBoost) classifiers to predict postoperative infection status at three timepoints: Before surgery (M1), end of surgery (M2) and at POD2 (M3). The model cohort with clear infection status was used to train the prediction model.

Preoperative and intraoperative factors are shown in Figure 1. All numeric input variables were scaled prior to training the classifiers. Preprocessing details and hyperparameter configurations are provided in Supplementary Table 7. To address class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) was applied within each training fold during training⁴⁸. Classifier performance was assessed using 5-fold cross-validation (CV), with evaluation metrics including sensitivity ($TP / (TP + FN)$), specificity ($TN / (TN + FP)$), Positive Predictive Value ($PPV = TP / (TP + FP)$), and the area under the receiver operating characteristic curve (AUROC). To quantify the incremental value of intraoperative features on the M2 model, we performed an ablation study comparing summary, trend, and higher-order statistics (HOS). Furthermore, the association between physiological dynamics and surgical duration was assessed using an ordinal logistic regression model adjusted for preoperative risk factors and procedure categories, with multiple testing correction using the Benjamini-Hochberg False Discovery Rate (FDR).

Model calibration

Model calibration was evaluated for each procedure cluster to assess agreement between out-of-fold predicted and observed infection risks. Calibration curves were generated by plotting the observed infection incidence against mean out-of-fold predicted probabilities within each cluster. Perfect calibration corresponds to the 45° line, where predicted risk equals observed risk. Points above the line indicate underestimation of risk, and points below indicate overestimation. To further assess overall accuracy, we calculated cluster-specific Brier scores and calibration bias. The Brier score reflects the average squared difference between predicted and actual outcomes. The score ranges from 0 (perfect prediction) to a maximum that depends on outcome incidence (0.25 when incidence $\approx 50\%$). Lower values indicate higher predictive accuracy. Calibration bias was calculated as the mean difference between predicted and observed infection incidence within each procedure cluster. Values near zero indicate good agreement. In combination, a low Brier score and small calibration bias supports reliable and generalizable risk estimation across the cohort.

Explainability of Model Predictions

Model explanations were generated by *post hoc* feature attribution using SHapley Additive exPlanations (SHAP)⁴⁹. We used the model-specific Tree Explainers to compute per-patient feature attributions and global importance was computed by mean absolute values. SHAP values were also plotted with coloured violin plots visualizing the distribution, direction, and magnitude of effects across patients. In order to improve visibility of the coloured violin plots, we removed values falling outside of the 99.9th quantile of the feature's SHAP values.

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Data Availability

The datasets used and analysed during the current study are not publicly available due to Swiss data protection regulations and institutional patient-privacy policies but are available upon reasonable request from the corresponding author.

Code Availability

The underlying code and model for this study is available on GitHub and can be accessed via this link: <https://github.com/T-Blatter/endofsurgeryinfection>.

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Author contributions

TUB led the conceptualization and development of the analyses, performed data processing, modelling, and validation work, and drafted the manuscript. YW contributed to the analysis and assisted with manuscript revision. OE contributed to conceptualization and data harmonization, modelling and analysis, coordinated resources and participated in manuscript revisions. KT contributed to study conceptualization and methodology, coordinated resources and manuscript editing. HG supported study design, contributed to the analysis, validation, and manuscript editing. GB contributed to conceptualization, oversight, visualization, and drafting and editing of the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing financial or non-financial interests.

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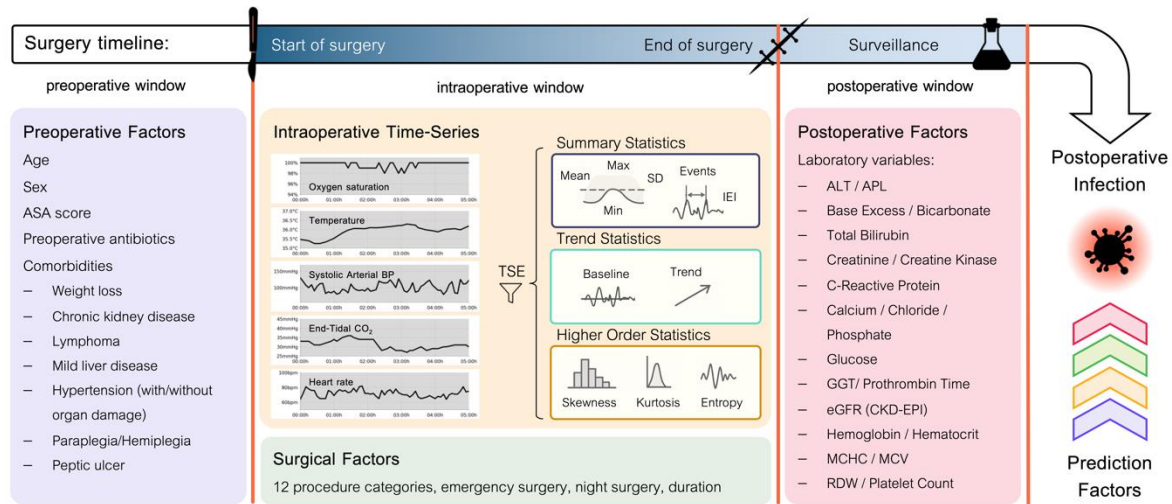


Figure 1. Framework for predicting postoperative infections using preoperative, intraoperative and postoperative variables. Preoperative factors consist of patient's variables, preoperative antibiotic use, and Charlson or Elixhauser comorbidities groups. The signals captured during the intraoperative window are summarized through time-series extraction (TSE), which include summary statistics (mean, standard deviation (SD), maximum, events [positive=peaks, negative=dips], and Inter-Event Intervals [IEI]), trend statistics (baseline, trend over time), and higher order statistics (skewness, kurtosis, entropy). Surgical factors include procedure category, emergency status, night surgery, duration of surgery. Postoperative variables consisted of laboratory values, collected at various postoperative days (0-2).

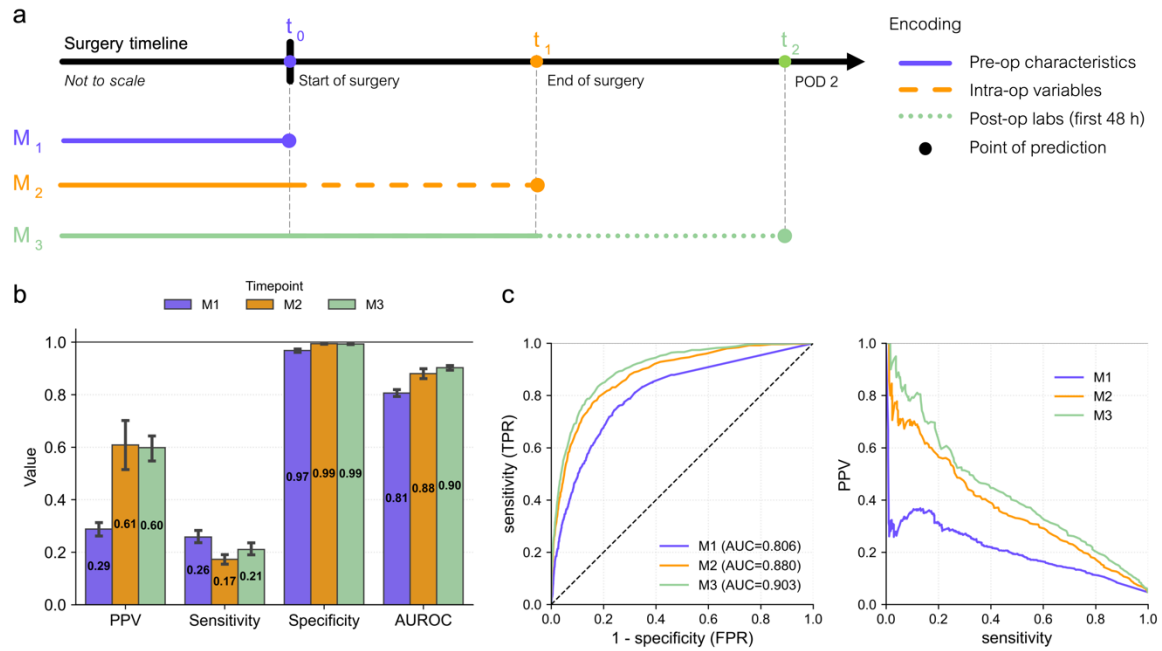


Figure 2. Model performance across the surgical trajectory. **a:** Three models were evaluated: M_1 (start of surgery): preoperative patient factors only. M_2 (end of surgery): preoperative plus intraoperative factors, and M_3 (POD2): preoperative plus intraoperative factors plus postoperative laboratory data on day 2. **b:** PPV, sensitivity, specificity, and AUROC metrics of the Random Forest model. **c:** Performance curves for the Random Forest model. Left: ROC curves displaying sensitivity vs. 1 – specificity. Right: Precision-Recall curves displaying positive predictive value (PPV) vs. sensitivity. Abbrev.: AUROC, area under the receiver operating characteristic; FPR, false positive rate; POD, postoperative day; PPV, positive predictive value; ROC, Receiver operating characteristic; TPR, true positive rate.

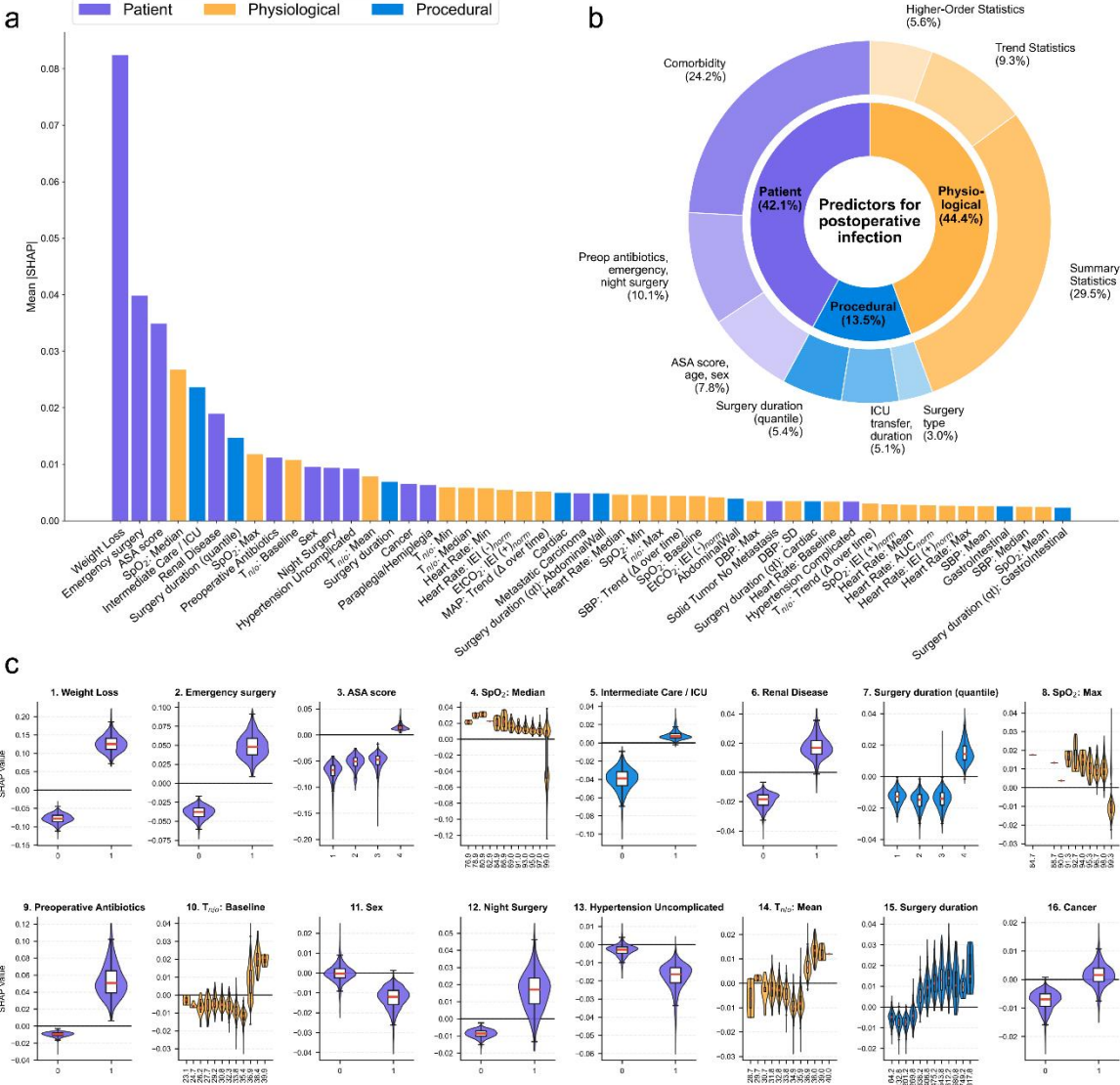


Figure 3. Feature importance and contribution of patient, procedural, and intraoperative physiological features to end-of-surgery infection predictions (M2). **a:** Ranking of the top 50 features based on mean absolute SHAP value and coloured by feature group (Patient, Physiological, Procedural). **b:** Nested decomposition of global feature importance based on summed mean absolute SHAP values. **c:** Grouped violin-plot graphs of the Top 16 ranked features showing the distribution of SHAP values across feature levels. The y-axis represents the SHAP value, where values >0 indicate an increased risk of infection and values <0 indicate a protective effect or lower risk relative to the cohort baseline. For binary variables, 0 and 1 represent the absence or presence/category of the trait. For continuous or ordinal variables, the x-axis reflects specific measurements or quantiles.

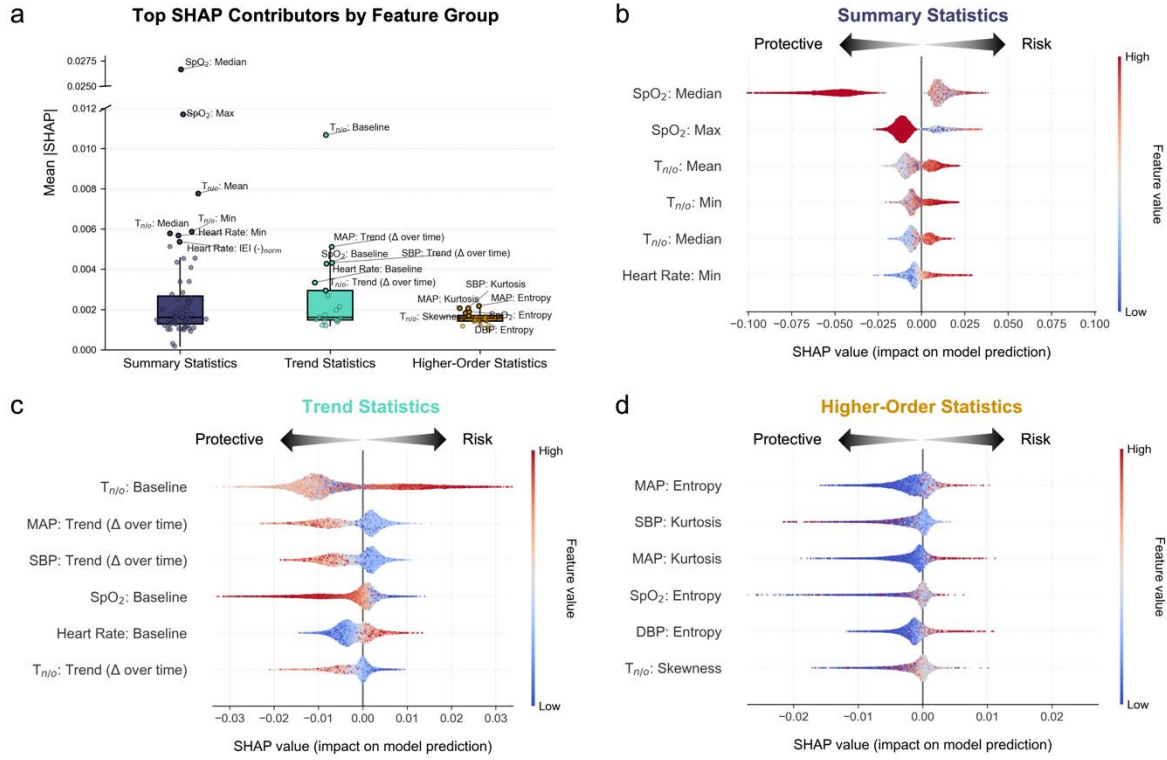


Figure 4. Contribution of time-series features to model prediction using SHAP analysis **a:** SHAP grouped summary plots of the different statistical time-series features. **b-d:** For each feature group, the model's prediction is visualized with the horizontal axis showing the magnitude and direction of a feature's contribution to the prediction (positive/negative), and the colour scale indicating the feature value (red = high, blue = low). Sorted by the model impact importance (absolute mean value). SHAP values represent the magnitude of each feature's impact on model predictions. Abbrev.: DBP, diastolic blood pressure, $EtCO_2$, end-tidal carbon dioxide; MAP, mean arterial blood pressure; SBP, systolic blood pressure; SpO_2 , peripheral oxygen saturation; $T_{n/o}$, nasopharyngeal temperature.